

Aim

To write a program to find the 1-norm, 2-norm and infinity norm of the matrix and display the result in two decimal places.

Equipment's required:

1. Hardware – PCs
2. Anaconda – Python 3.7 Installation / Moodle-Code Runner

Algorithm:

1. Get the input matrix using `np.array()`
2. Find the 2-norm of the matrix using `np.linalg.norm()`
3. Print the norm of the matrix in two decimal places.

Program:

```
# Register No: 212225220122
# Developed By: vishwajith p
# 1-Norm of a Matrix

import os
os.environ["OPENBLAS_NUM_THREADS"]="1"
import numpy as np
matrix=eval(input())
one_matrix=np.linalg.norm(matrix,1)
print("{:.2f}".format(one_matrix))

# 2-Norm of a Matrix
import os
os.environ["OPENBLAS_NUM_THREADS"]="1"
import numpy as np
matrix=eval(input())
two_matrix=np.linalg.norm(matrix,2)
print("{:.2f}".format(two_matrix))

# Infinity Norm of a Matrix
```

```
import os
os.environ["OPENBLAS_NUM_THREADS"]="1"
import numpy as np
matrix=eval(input())
inf_matrix=np.linalg.norm(matrix,np.inf)
print("{:.2f}".format(inf_matrix))
```

Output:

1-Norm of a Matrix

The screenshot shows a web browser window with the URL <https://lms2.ai.saveetha.in/mod/quiz/review.php?attempt=1339963&cmid=2677>. The page displays the following content:

```
1 #Program to find the 1-Norm of a matrix and display the results in two decimal places.
2 #Developed by: VISHWAJITH P
3 #RegisterNumber:212225220122
4 import os
5 os.environ["OPENBLAS_NUM_THREADS"]="1"
6 import numpy as np
7 matrix=eval(input())
8 one_matrix=np.linalg.norm(matrix,1)
9 print("{:.2f}".format(one_matrix))
```

	Input	Expected	Got	
✓	<code>[[-1, 3],[3, -4],[1, 7]]</code>	14.00	14.00	✓
✓	<code>[[1, 2, 3],[-3,-4,-1],[9,6,1]]</code>	13.00	13.00	✓

Passed all tests! ✓

▶ Show/hide question author's solution (Python3)

Correct

2-Norm of a Matrix

SEC

```

2 Program to find 2-norm of a matrix.
3 Developed by: VISHWAJITH P
4 RegisterNumber: 212225220122
5 ...
6 import os
7 os.environ["OPENBLAS_NUM_THREADS"]="1"
8 import numpy as np
9 matrix=eval(input())
10 two_matrix=np.linalg.norm(matrix,2)
11 print("{:.2f}".format(two_matrix))

```

	Input	Expected	Got	
✓	[[1,2],[3,4]]	5.46	5.46	✓
✓	[[-1, 3],[3, -4],[1, 7]]	8.66	8.66	✓
✓	[[2, 3],[3, 4],[1, 8]]	9.86	9.86	✓

Passed all tests! ✓

► Show/hide question author's solution (Python3)

Infinity Norm of a Matrix

SEC

```

2 Program to find Infinity of a matrix.
3 Developed by: VISHWAJITH P
4 RegisterNumber: 212225220122
5 ...
6 import os
7 os.environ["OPENBLAS_NUM_THREADS"]="1"
8 import numpy as np
9 matrix=eval(input())
10 inf_matrix=np.linalg.norm(matrix,np.inf)
11 print("{:.2f}".format(inf_matrix))

```

	Input	Expected	Got	
✓	[[-1, 3],[3, -4],[1, 7]]	8.00	8.00	✓
✓	[[1,2,3],[-9,-8,-3],[10,3,2]]	20.00	20.00	✓

Passed all tests! ✓

► Show/hide question author's solution (Python3)

Correct

Marks for this submission: 1.00/1.00.

Result

Thus the program for 1-norm, 2-norm and Infinity norm of a matrix are written and verified.